Trishit Mukherjee

Mathematically grounded deep learning for inverse imaging and stable reconstruction

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Research Profile

Junior Research Fellow at the Indian Institute of Science working at the intersection of inverse problems, computational imaging, operator learning, and optimization theory. My work develops expressive neural denoisers and learned regularizers/ priors with explicit *convergence* guarantees for iterative reconstruction, together with GPU-accelerated implementations in PyTorch, C++, and CUDA. I am a coauthor of papers accepted to ECCV'26 and ICML'26, and I am seeking a PhD in inverse problems, computational imaging, theoretically grounded computer vision, graph neural networks for signal and image processing domains.

Education

M.Tech. in Artificial Intelligence, Indian Institute of Science (IISc), Bengaluru 2023–2025
CGPA: 8.4/10

B.E. in Computer Science and Engineering, Jadavpur University, Kolkata 2019–2023
CGPA: 8.81/10

Research Projects and Publications

Contractive Networks for Stable Image Reconstruction (ECCV 2026) *May'24–Jun'25*
Master's Thesis · Prof. Kunal N. Chaudhury, EECS, IISc

- "**Stabilizing Deep Reconstruction Operators with Contractive Anchoring**". [arXiv | code]
  Arghya Sinha, Trishit Mukherjee, and Kunal N. Chaudhury. *European Conference on Computer Vision (ECCV), 2026*.
- "**ViSTA: Viscosity Stabilized Plug-and-Play Reconstruction**". Arghya Sinha, Trishit Mukherjee, and Kunal N. Chaudhury. *arXiv:2508.01441*, 2025.
- Formalized the *peak-and-collapse* instability of iterative reconstruction via a local *stability index* η that measures the expansiveness of any black-box PnP/RED operator, and built an adaptive *contractive-anchoring* stabilizer that provably controls η as a drop-in step – with no retraining or modification of the pretrained denoiser.
- Designed trainable contractive gradient-step CNN denoisers (spectrally controlled linear maps with learnable spline nonlinearities) that serve as strong, lightweight anchors; validated across algorithmic solvers and modern deep denoisers on deblurring, superresolution and MRI, sustaining 0.5–1 dB PSNR gains across iterations.
  Stack: *PyTorch*, `DeepInverse`, *OpenCV, scikit-image, NumPy, Matplotlib*

Trainable Kernel Denoisers for Convergent Image Reconstruction (ICML 2026) *Dec'25–Present*
Prof. Kunal N. Chaudhury, EECS, IISc

- "**Trainable Nonexpansive Denoisers for Contractive Image Reconstruction**". [arXiv | code]
  A. Sinha, A. Banerjee, T. Mukherjee, and K. N. Chaudhury. *International Conference on Machine Learning (ICML), 2026*.
- Proposed a permutation-based weighted-aggregation denoiser whose global *nonexpansiveness* is guaranteed by construction (permutation group action + a symmetrization protocol), using a compact weight predictor.
- Proved that the induced PnP-HQS reconstruction map is *contractive* under mild conditions, giving global convergence, and matched the unconstrained DRUNet on denoising while remaining provably stable on deblurring and superresolution.
  Stack: *PyTorch*, `DeepInverse`, *OpenCV, scikit-image, NumPy, Matplotlib*

`neural_shift_cuda`: Parallel Kernel Aggregation through CUDA acceleration *July'26–Present*
Open-source C++/CUDA extension (lead developer)

- The reference implementation of Kernel Denoisers (ICML 2026) processed all image shifts one at a time in Python, turning a naturally parallel operation into a serial bottleneck. The project exposes that parallelism with fused C++/CUDA kernels that gather shifted pairs and accumulate the stochastic denoiser update in a single GPU pipeline.
- Built as drop-in patches: the accelerated path preserves the original model, parameters, and operator while replacing only the slow execution route. It includes a pure-PyTorch CPU fallback, full autograd, and reference-equivalence tests (bit-exact in `fp64`, $\sim$`1e-5` in `fp32`), with Pascal–Blackwell (`sm_120`) support.
- **Benchmarks**: Full-step speedups of **2.23**$\times$ and **1.31**$\times$ for the Kernel Denoisers with an improved UNet based feature extraction– attentive weight predictor architecture with symmetric and asymmetric kernels respectively, and **1.61**$\times$ and **1.33**$\times$ for the original NECTR implementations.
  Stack: *CUDA, C++, PyTorch (custom C++/CUDA extensions)*, `nvcc`, *pytest*

Location Metadata Extraction from Geosocial Road-Accident Data *Aug'22–May'23*

Research Project · Prof. Sarbani Roy, Dept. of CSE, Jadavpur University

- "**Location Metadata Extraction from Geosocial Data of Road Accidents using Deep Learning Models**". Trishit Mukherjee, Soumitra Sinhahajari, Debargha Mukherjee, Hrishikesh Mallick, Asif Iqbal Middya, and Sarbani Roy. *Evolving Systems*, 16(1), 2025.
- Built an information-extraction pipeline for road-accident news: API-based collection, LSTM/Word2Vec relevance classification (96% accuracy, 0.94 F1), and BERT-based location extraction (0.673 exact match); findings published in *Evolving Systems*.
  Stack: *TensorFlow, PyTorch, Transformers, MongoDB, Flask*

Work Experience & Teaching

Junior Research Fellow – LISA, EE IISc *Aug'25–Present*

Supervisor: Prof. Kunal N. Chaudhury · Bengaluru, India

- Developing trainable stochastic denoisers and graph / kernel-Laplacian regularizers for image reconstruction – learned edge weights, PD reconstruction systems, conjugate-gradient solves, and implicit differentiation – alongside the ECCV 2026 and ICML 2026 work above.

Teaching Assistant – Linear Algebra & Applications, EECS IISc *Aug–Nov'24*

Instructor: Prof. Kunal N. Chaudhury · Bengaluru, India

- Ran tutorials and helped develop assignment material for undergraduate and graduate students for the IISc Mtech AI core course.

Software Engineering Intern – Innovation Junction LLC *Jun–Aug'21*

Software Engineering Intern · Remote, USA

- Built AWS data pipelines: CRUD operations on CSV data, SQL/NoSQL ingestion into EC2, and Power BI reporting.

Technical Skills & Coursework

Research: Inverse problems; computational imaging; PnP/RED; fixed-point & operator theory; contractive/nonexpansive operators; graph signal processing; learned regularization
Programming: Python, C++, CUDA C, C, Java, JavaScript, LaTeX
ML & Imaging: PyTorch, DeepInverse, TensorFlow, Hugging Face Transformers, NumPy, SciPy, scikit-image, scikit-learn, OpenCV
Systems & Tools: Linux, Git/GitHub, `nvcc`, CUDA extensions, Slurm, Jupyter, tmux, MongoDB, MySQL, AWS
Relevant Coursework: Linear Algebra & Applications; Computational Methods of Optimization; Advanced Convex Optimization; Advanced Image Processing; Pattern Recognition & Neural Networks; Stochastic Models & Applications

Achievements

- **WBJEE 2019:** All India Rank 139.
- **GATE 2023 (Computer Science and Information Technology):** All India Rank 39.
- **Junior Research Fellow, FSID, IISc:** Supported by the PGCoE project "Multilevel Optimization for Power Grid Security and Defense" (Project No. `FSID/25-26/PC00006T`).
- **Junior Research Fellow, IISc:** Supported by the Anusandhan National Research Foundation (ANRF), Government of India, under Grant No. `ANRF/ARG/2025/00696/ENS`.